

George Whittle

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Objective statement

Looking for quantitative research intern roles at a top quantitative hedge fund. Significant research experience in an academic and professional quantitative setting, with a particular interest in furthering my experience in a financial setting. Skilled coder able to rapidly prototype from research with experience developing and deploying large projects. Special interests in Bayesian machine learning (meta-learning, approximate inference, Bayesian sensor fusion, probabilistic numerical methods i.e. Bayesian optimisation and quadrature), optimisation (convex and non-convex global) and stochastic systems. Current research interests in fundamental Neural Network theory (Neural Tangent Kernel theory and Bayesian interpretations of feature learning).

Education

DPhil Engineering Science – Machine Learning Research Group, University of Oxford 2024 – 2027

- Project title: Dynamic Bayesian Machine Learning for Quantum Device Control
- Bayesian Exploration Lab and Ares Lab (co-supervised by Professors Michael Osborne and Natalia Ares)
- Main project developing novel algorithms capable of modelling and tracking disorder potential fields in silicon qubits
- Papers on Amortised Bayesian Inference for generalised Bayesian filtering tasks, and Neural Tangent Kernel theory

MEng Engineering Science – Somerville College, University of Oxford 2020 – 2024

- First class, ranked 3rd out of 141 overall with 2nd out of 176 in 2nd year exams
- Specialised in Machine Learning, Optimisation, Stochastic Control and Mathematical Techniques
- Awarded Edgell Sheppee prize by the Department of Engineering Science for excellent performance, Principal's prize and Mary Somerville prize by Somerville College for performance in 4th year exams
- Academic Scholar (Awarded by Somerville College for performance in 1st, 2nd and 3rd year exams)

Quantitative Finance Experience

Quant Research Internship – Point72 (15 Weeks) June 2025 - Present

- Systematic alpha research for L/S equities pod, focusing on flow and regime analysis
- High Sharpe Ratio statistical arbitrage strategies performant when other signals in book in drawdown

Quant Portfolio Manager – Oxford Alpha Fund January 2024 - Present

- Led team of 5 researching the use of Bayesian quadrature in robust systematic strategy ensemble construction
- Now Head of Research on the Executive Committee, overseeing quant research projects and delivering bootcamp lectures

Quant Research Internship – Gresham Quant (10 Weeks) Summer 2023

- Worked in research team of multi award winning systematic Hedge Fund (best CTA globally 2021)
- Completely redesigned portfolio optimisation algorithm, researching and implementing a liquidity-aware dynamic rebalancing algorithm based on hierarchical constrained convex optimisation
- Carried out alpha research into optimisation of contract roll trade timing, delivering a total book back test net Sharpe Ratio improvement of 0.07 in the ideal case and after consulting with traders, 0.05 in the realistic case

Other Relevant Experience

Machine Learning Scientist (Defence) – Mind Foundry August 2024 - Present

- Part time collaboration during DPhil researching transformer-based online Bayesian filtering
- Previously a Machine Learning Scientist intern researching high dimensional nonconvex multi-objective optimisation

Optimisation Consultant – Lexington Medical

February 2024 - Present

- Designed and implemented statistical optimisation software performing probabilistic demand modelling and forecasting, and optimisation of global inventory and shipping configuration of FG to 3PL network

Lecturer (SCR) – Somerville College, University of Oxford

October 2024 - Present

- First- and second-year electronics, control theory, and communications theory
- Inter-college classes on control theory, communications theory, and algorithms and data structures

4th Year Project – University of Oxford

October 2023 – June 2024

- Project title: LLM Enhanced Bayesian Quadrature for Neural Ensemble Search
- Used LLMs to analyse source code of candidate models and produce a common coordinate space, over which a Gaussian process prior can be defined for use in Bayesian quadrature, producing ensemble weights which efficiently balance model performance and diversity
- Developed a large, well-structured and documented code repository supporting the project
- Outperformed state-of-the-art approaches on benchmarks, and performed well on tasks current approaches cannot tackle

EUROP Research Internship – University of Oxford (8 Weeks)

Summer 2022

- Control group, Department of Engineering Science
- Researched Neural Kalman Filters, developing novel training algorithm based on trajectory simulation

Software Development Skills

- 12 years' experience in Python across various paradigms, and recent experience in MATLAB and C++
- Formally educated in software engineering, object-oriented programming, algorithms and data structures, design patterns
- Experience in managing large software repositories using Git
- Experience interacting with databases using SQL
- Particularly familiar with Numpy, Scipy, Pandas and PyTorch libraries
- Developed python libraries for various numerical methods including pseudo-spectral differentiators, implicit integrators and probabilistic numerical methods including Bayesian optimisation and quadrature
- Experience developing large, scalable software professionally
- Received an internship offer for a Quant Developer role at Beacon Platform

Other Experience and Interests

Competitive Experience

- Reserve on British team for (cancelled) 2020 International Physics Olympiad (top 6 in country)
- Top gold in UK Physics Olympiad (top 100 in country)
- Golds in UK Chemistry Olympiad, Cambridge Chemistry Challenge and 7 consecutive UKMT maths challenges

Teaching

- Tutored an MEng Engineering Science undergraduate on first- to third-year mathematics July 2023-Present
- Tutored Oxbridge engineering admissions with Swift Education March 2024-Present

Committees

- Head of Research, Executive Committee, Oxford Alpha Fund October 2024-Present
- VP of Sponsors and Events, Oxford University Engineering Society October 2023-May 2024

Music – Working towards LTCL diploma in Piano, grade 8 in Trumpet, former First Trumpet of Oxford Millenium orchestra, various college, and inter-college ensembles